Z10.5 (cont)

TM5-DSBC **Report on THICKNESS OF WATERTIGHT TRANSVERSE BULKHEADS IN CARGO HOLDS** **Sheet 9**

Ship's name..... Class Identity No. ..... Report No. .....

LOCATION OF STRUCTURE: FRAME NO.:

STRUCTURAL COMPONENT (PLATING/STIFFENER)

	Original Thickness mm	Max. Alwb. Dim. mm	Gauged		Diminution P		Diminution S	
			Port	Starboard	mm	%	mm	%

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM5-DSBC

1. This report form is to be used for recording the thickness measurement of:

   W.T. transverse bulkheads in cargo holds

2. Guidance for areas of measurements is indicated on sheet 14 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

4. The maximum allowable diminution could be stated in an attached document.

Z10.5 (cont) **TM6-DSBC** **Report on THICKNESS MEASUREMENT OF MISCELLANEOUS STRUCTURAL MEMBERS** **Sheet 10**

Ship's name..... Class Identity No. ..... Report No. .....

STRUCTURAL MEMBER:									SKETCH
LOCATION OF STRUCTURE:									
Description	Org. Thk. mm	Max. Alwb. Dim. mm	Gauged		Diminution P		Diminution S		
			P	S	mm	%	mm	%	

Operators Signature..... NOTES – See Reverse

Z10.5
(cont)

NOTES TO REPORT TM6-DSBC

1. This report is to be used for recording the thickness measurement of:

   Miscellaneous structural members including the structural items (40), (41) and (42) as shown on diagrams of typical transverse sections illustrated on sheets 11, 12 and 13 of this document.

2. Guidance for areas of measurement is indicated on sheet 14 of this document.

3. The single measurements recorded are to represent the average of multiple measurements.

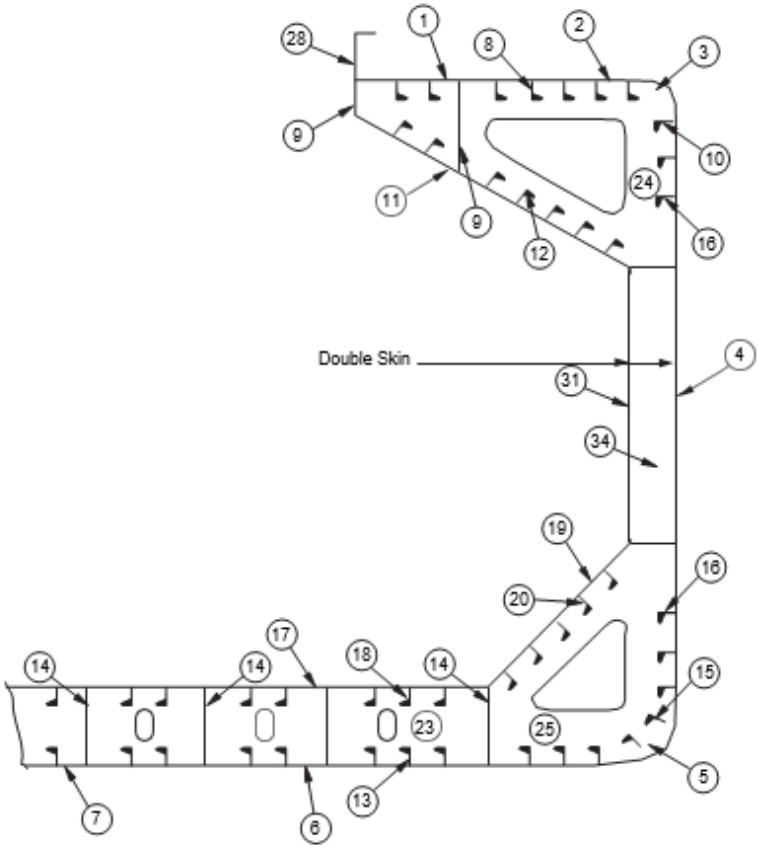
4. The maximum allowable diminution could be stated in an attached document.

Sheet 11

Z10.5 (cont)

Thickness Measurement - Double Skin Bulk Carriers

Typical transverse section of a double skin bulk carrier with indication of longitudinal and transverse members



Report on TM2-DSBC (i) & (ii)	
1.	Strength deck plating
2.	Stringer plate
3.	Sheerstrake
4.	Side shell plating
5.	Bilge plating
6.	Bottom shell plating
7.	Keel plate

Report on TM3-DSBC			
8.	Deck longitudinals	17.	Inner bottom plating
9.	Deck girders	18.	Inner bottom longitudinals
10.	Sheerstrake longitudinals	19.	Hopper plating
11.	Topside tank sloping plate	20.	Hopper longitudinals
12.	Topside tank sloping plate longitudinals	31.	Inner side plating
13.	Bottom longitudinals	-	Inner side longitudinals, if any
14.	Bottom girders	-	Horizontal girders in wing ballast tanks
15.	Bilge longitudinals		
16.	Side shell longitudinals, if any		

Report on TM4-DSBC	
23.	Double bottom tank floors
25.	Hopper side tank transverses
34.	Transverse web frame
-	Topside tank transverses

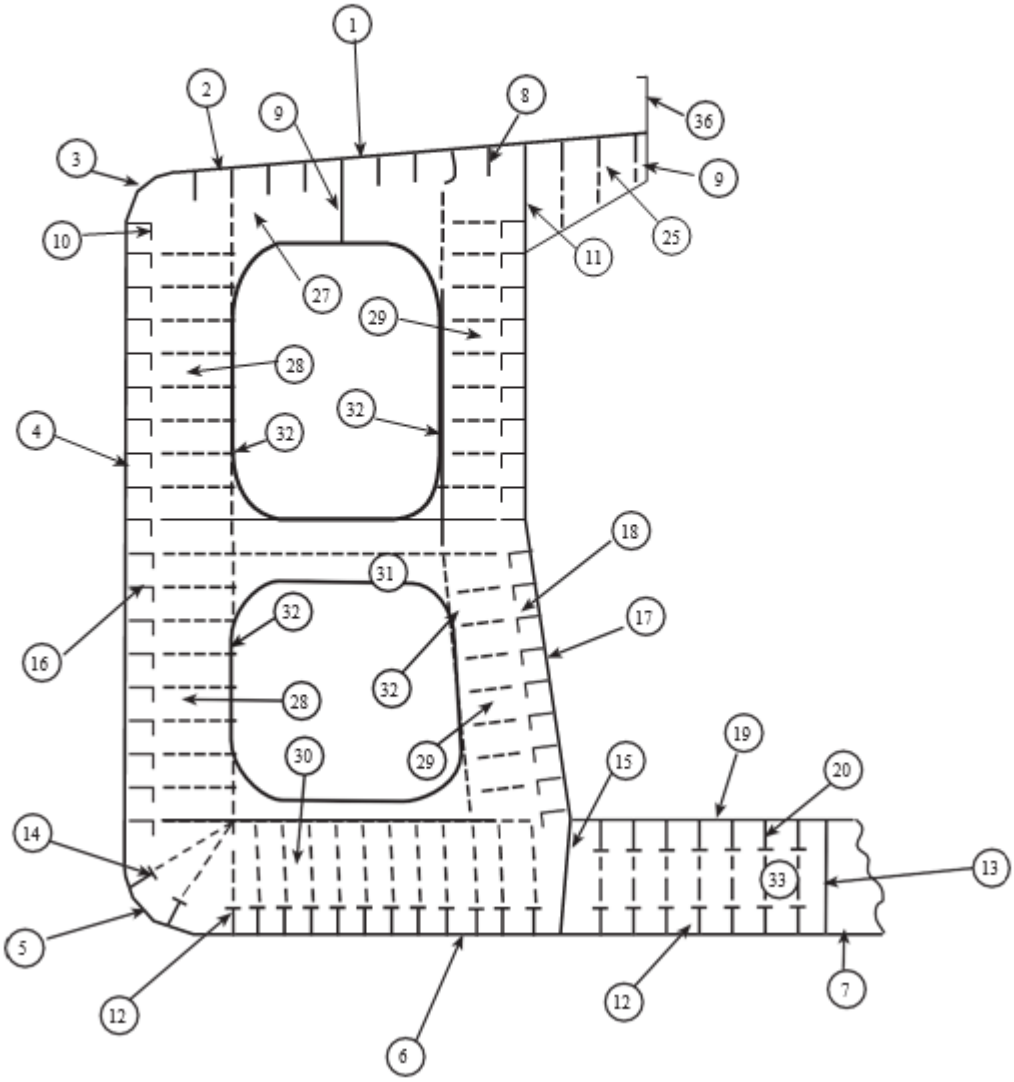
Report on TM6-DSBC	
28.	Hatch coamings
-	Deck plating between hatches
-	Hatch covers

Sheet 12

Z10.5 (cont)

Thickness Measurement - Ore Carriers

Typical transverse section of an ore carrier with indication of longitudinal and transverse members



Report on TM2-DSBC (i) & (ii)	
1.	Strength deck plating
2.	Stringer plate
3.	Sheerstrake
4.	Side shell plating
5.	Bilge plating
6.	Bottom shell plating
7.	Keel plate

Report on TM3-DSBC	
8.	Deck longitudinals
9.	Deck girders
10.	Sheerstrake longitudinals
11.	Longitudinal bulkhead top strake
12.	Bottom longitudinals
13.	Bottom girders
14.	Bilge longitudinals
15.	Longitudinal bulkhead lower strake
16.	Side shell longitudinals
17.	Longitudinal bulkhead plating (remainder)
18.	Longitudinal bulkhead longitudinals
19.	Inner bottom plating
20.	Inner bottom longitudinals
21.	
22.	
23.	
24.	

Report on TM4-DSBC	
25.	Deck transverse centre tank
26.	Bottom transverse centre tank
27.	Deck transverse wing tank
28.	Side shell vertical web
29.	Longitudinal bulkhead vertical web
30.	Bottom transverse wing tank
31.	Struts
32.	Transverse web face plate
33.	D.B. Floors
34.	
35.	

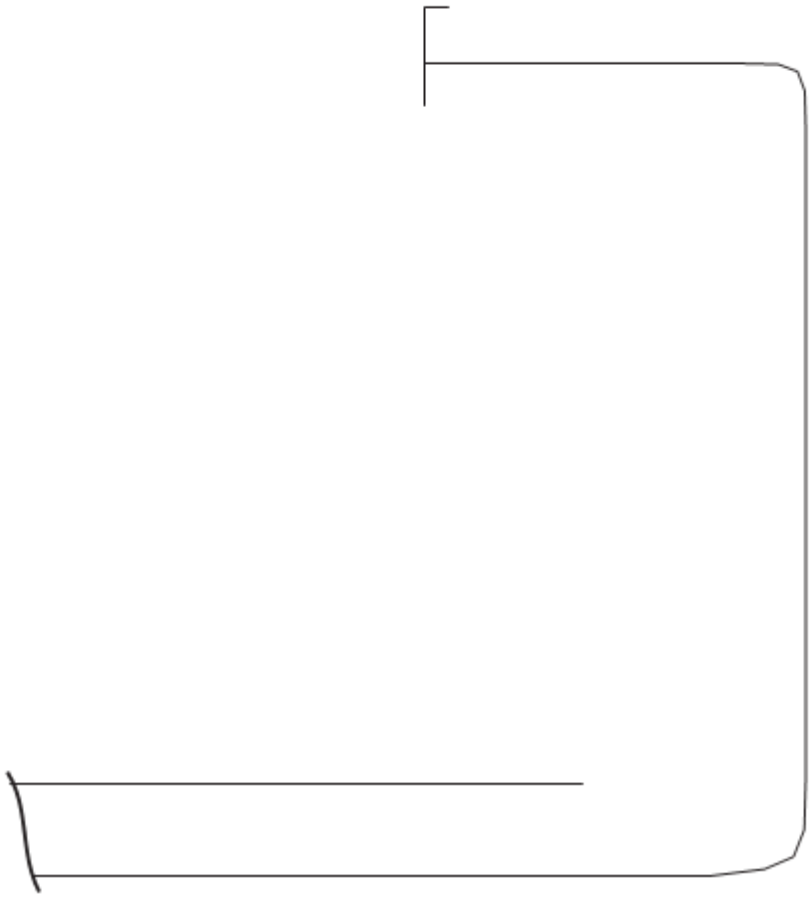
Report on TM6-DSBC	
36.	Hatch coamings
37.	Deck plating between hatches
38.	Hatch covers
39.	
40.	

Z10.5
(cont)

Sheet 13

Thickness Measurement - Double Skin Bulk Carriers

Transverse section outline: The diagram may be used for those ships where the diagrams on sheet 11 and 12 are not suitable



Report on TM2-DSBC (i) & (ii)	
1.	Strength deck plating
2.	Stringer plate
3.	Sheerstrake
4.	Side shell plating
5.	Bilge plating
6.	Bottom shell plating
7.	Keel plate

Report on TM3-DSBC			
8.	Deck longitudinals	17.	Inner bottom plating
9.	Deck girders	18.	Inner bottom longitudinals
10.	Sheerstrake longitudinals	19.	Hopper plating
11.	Topside tank sloping plate	20.	Hopper longitudinals
12.	Topside tank sloping plate longitudinals	31.	Inner side plating
13.	Bottom longitudinals	-	Inner side longitudinals, if any
14.	Bottom girders	-	Horizontal girders in wing ballast tanks
15.	Bilge longitudinals		
16.	Side shell longitudinals, if any		

Report on TM4-DSBC	
23.	Double bottom tank floors
25.	Hopper side tank transverses
34.	Transverse web frame
-	Topside tank transverses

Report on TM6-DSBC	
28.	Hatch coamings
-	Deck plating between hatches
-	Hatch covers

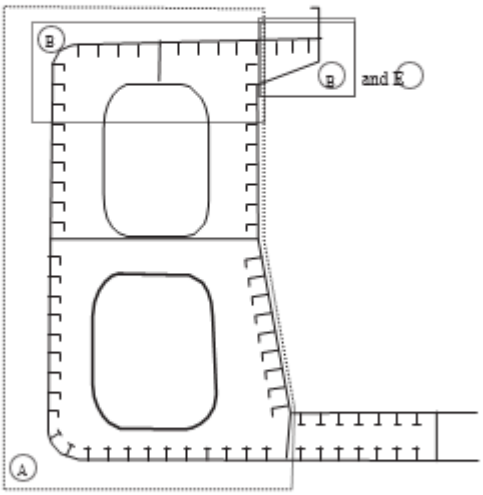
Z10.5
(cont)

Sheet 14

Thickness Measurement - Ore Carriers

Areas subject to close up survey and thickness measurements - areas (A) to (E) as defined in Table I of UR Z10.5 - Thickness to be reported on TM3-DSBC, TM4-DSBC, TM5-DSBC and TM6-DSBC as appropriate

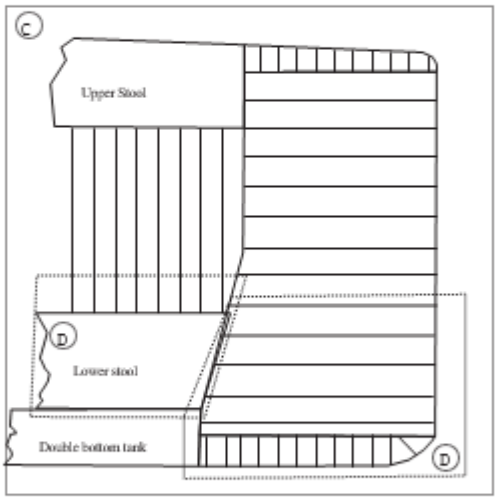
Typical transverse section close-up survey



Thickness to be reported on TM3-DSBC and TM4-DSBC as appropriate

Close-up survey area

Typical transverse bulkhead



Thickness to be reported on TM5-DSBC

Z10.5
(cont)

ANNEX II (CSR)

IACS RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS OF DOUBLE SKIN BULK CARRIERS BUILT UNDER IACS COMMON STRUCTURAL RULES*

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*
Note: Annex II (CSR) is recommendatory.

Z10.5
(cont)

Sheet 1

CONTENTS

Sheet 1	-	Contents
Sheet 2	-	Instructions
Sheet 3	-	General particulars

REPORTS

Sheet 4	-	Report TM1-DSBC(CSR) for recording the thickness measurement of all deck plating, all bottom plating and side shell plating
Sheet 5	-	Report TM2-DSBC(CSR) (i) for recording the thickness measurement of shell and deck plating at transverse sections - strength deck and sheerstrake plating
Sheet 6	-	Report TM2-DSBC(CSR) (ii) for recording the thickness measurement of shell plating at transverse sections
Sheet 7	-	Report TM3-DSBC(CSR) for recording the thickness measurement of longitudinal members at transverse sections (including double hull structure)
Sheet 8	-	Report TM4-DSBC(CSR) for recording the thickness measurement of transverse structural members (including common frames, web frames and transverse bulkheads in double hull tanks)
Sheet 9	-	Report TM5-DSBC(CSR) for recording the thickness measurement of W.T. transverse bulkheads in cargo holds
Sheet 10	-	Report TM6-DSBC(CSR) for recording the thickness measurement of miscellaneous structural members

GUIDANCE

Sheet 11	-	Typical transverse section of a double skin bulk carrier. The diagram includes details of the items to be measured and the report forms to be used.
Sheet 12	-	Thickness Measurement - Double Skin Bulk Carriers Transverse section outline. This diagram may be used for those ships where the diagram on sheet 11 is not suitable.
Sheet 13	-	Sketches of double skin bulk carrier showing typical areas for thickness measurement of structural members and transverse bulkheads in association with close-up survey requirements. – areas (A) to (E) as defined in Table I of the UR Z10.5.

Z10.5
(cont)

INSTRUCTIONS

Sheet 2

IACS Recommended Procedures for Thickness Measurements of Double Skin Bulk Carriers Built under IACS Common Structural Rules

1. This document is to be used for recording thickness measurements of double skin bulk carriers built under IACS Common Structural Rules as required by the IACS Unified Requirement Z10.5.

2. Reporting forms TM1-DSBC(CSR), TM2-DSBC(CSR) (i) and (ii), TM3-DSBC(CSR), TM4-DSBC(CSR), TM5-DSBC(CSR) and TM6-DSBC(CSR) (sheets 4-10) are to be used for recording thickness measurements. The as-built thickness and the voluntary thickness addition and renewal thickness (minimum allowable thickness) are to be stated in the said forms.

3. The remaining sheets 11-13 are guidance diagrams and notes relating to the reporting forms and the procedure for the thickness measurement.

Z10.5 (cont)

GENERAL PARTICULARS

Sheet 3

Ships name:-

IMO number:-

Class identity number:-

Port of registry:-

Gross tons:-

Deadweight:-

Date of build:-

Classification Society:-

Name of Firm performing thickness measurement:-

Thickness measurement firm certified by:-

Certificate No:-

Certificate valid from.....to.....

Place of measurement:-

First date of measurement:-

Last date of measurement:-

Special survey/intermediate survey due:-*

Details of measurement equipment:-

Qualification of operators:-

Report Number:- consisting of.....Sheets

Name of operator:- ..... Name of surveyor:- .....

Signature of operator:-. ..... Signature of surveyor:- .....

Firm official stamp:- Classification Society Official Stamp:-

* Delete as appropriate

Z10.5 (cont) **TM1-DSBC(CSR)**

Report on THICKNESS MEASUREMENT of ALL DECK PLATING, ALL BOTTOM PLATING or SIDE SHELL PLATING* (* - delete as appropriate)

Sheet 4

Ship's name..... Class Identity No..... Report No.....

STRAKE POSITION																		
PLATE POSITION	No. or Letter	As Built Thk. mm	Voluntary Thickness Addition mm	Renewal Thickness mm (a)	Forward Reading						Aft Reading						Mean Remaining Corr. Addition, mm [(c1)+(c2)]/2	
					Gauged Thk. mm (b1)		Remaining Corr. Addition, mm (c1)=(b1)-(a)				Gauged Thk. mm (b2)		Remaining Corr. Addition, mm (c2)=(b2)-(a)					
					P	S	P		S		P	S	P		S		P	S
12th forward																		
11th																		
10th																		
9th																		
8th																		
7th																		
6th																		
5th																		
4th																		
3rd																		
2nd																		
1st																		
Amidships																		
1st aft																		
2nd																		
3rd																		
4th																		
5th																		
6th																		
7th																		
8th																		
9th																		
10th																		
11th																		
12th																		

Operators Signature..... NOTES – See Reverse

Z10.5 (cont)

NOTES TO REPORT TM1-DSBC(CSR)

1. This report is to be used for recording the thickness measurement of:-

   1.1 All strength deck plating within cargo length area.

   1.2 All keel, bottom shell plating and bilge plating within the cargo length area.

   1.3 Side shell plating including selected wind and water strakes outside the cargo length area.

   1.4 All wind and water strakes within the cargo length area.

2. The strake position is to be cleared indicated as follows:-

   2.1 For strength deck indicate the number of the strake of plating inboard from the stringer plate.

   2.2 For bottom plating indicate the number of the strake of plating outboard from the keel plate.

   2.3 For side shell plating give number of the strake of plating sheerstrake and letter as shown on shell expansion.

3. Only the deck plating strakes outside line of openings are to be recorded.

4. Measurements are to be taken at the forward and aft areas of all plates and the single measurements recorded are to represent the average of multiple measurements.

5. The remaining corrosion addition is to be recorded with result of gauged thickness minus renewal thickness. If the result is negative, the structure in way shall be renewed, and the mark “R” is to be indicated in the right-hand column. If the result is between 0 and 0.5 mm (0 included), the structure in way shall be additional gauged, and the mark “S” is to be indicated in the right-hand column.

Z10.5 (cont) **TM2-DSBC(CSR) (i)** **Report on THICKNESS MEASUREMENT OF SHELL AND DECK PLATING (one, two or three transverse sections)**

Sheet 5

Ship's name..... Class Identity No..... Report No.....

STRENGTH DECK AND SHEERSTRAKE PLATING

	FIRST TRANSVERSE SECTION AT FRAME NUMBER										SECOND TRANSVERSE SECTION AT FRAME NUMBER										THIRD TRANSVERSE SECTION AT FRAME NUMBER									
STRAKE POSITION	No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)				No. or Letter	As Built Thk. mm	Vol. Thk. Add. mm	Ren. Thk. mm (a)	Gauged Thk. mm (b)		Remaining Corr. Addition, mm (b)-(a)			
					P	S	P		S						P	S	P		S						P	S	P		S	
Stringer Plate																														
1st strake inboard																														
2nd																														
3rd																														
4th																														
5th																														
6th																														
7th																														
8th																														
9th																														
10th																														
11th																														
12th																														
13th																														
14th																														
centre strake																														
sheer strake																														
TOPSIDE TOTAL																														

Operators Signature..... NOTES – See Reverse